

Multiple Choice (10 marks)

1. There are 72 students on the dance team, and 4 over 9 take tap lessons. How many students take tap lessons?
 - (a) 32
 - (b) 45
 - (c) 40
 - (d) 49
 - (e) 63
2. A tournament has 393 players with 3 players on each team. Which expression shows how to find the number of teams in the tournament?
 - (a) $3 + 393$
 - (b) 3×393
 - (c) $(393 - 3) \times 3$
 - (d) $393 \div 3$
 - (e) $(393 + 3) \div 3$
3. Coloring the edges of a complete graph with n vertices in 2 colors (red and blue), what is the smallest n that guarantees there is either a 4-clique in red or a 4-clique in blue?
 - (a) 12
 - (b) 15
 - (c) 14
 - (d) 16
 - (e) 22
4. The quadratic $2x^2 + 5x + 12 = 19 - 7x$ has two solutions. What is the positive difference between these solutions?
 - (a) $2\sqrt{3}$
 - (b) $\sqrt{7}$
 - (c) $3\sqrt{2}$
 - (d) $5\sqrt{2}$
 - (e) 2
5. What is the measure, in degrees, of an angle that is equivalent to $1/360$ of a circle?

- (a) 0
- (b) 720
- (c) 30
- (d) 180
- (e) 90

True / False (10 marks)

Answer True or False

6. True or False: The value of the series $\sum_{n=0}^{\infty} (-1)^n \frac{1}{3n+1}$ is approximately 0.935649.
7. True or False: The number must have 25 as a factor, as 25 is a product of its prime factors.
8. True or False: The correct answer to 'Suppose $x-3$ and $y+3$ are multiples of 7. What is the smallest positive integer, n , for which $x^2 + xy + y^2 + n$ is a multiple of 7?' is '2'.
9. True or False: The difference in distance walked in the morning and afternoon is 5 over 12 miles.
10. True or False: The value of p that satisfies the equation $24 = 2^p$ is $p = 6$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Tim wants to invest some money in a bank which compounds quarterly with an annual interest rate of 7%. To the nearest dollar, how much money should he invest if he wants a total of \$60,000 at the end of 5\$years?
11. Ms. Gutierrez is preparing for her gym class of 32 students, each requiring a rope piece of 5 feet 8 inches. Calculate the total length of rope needed and discuss the implications of unit conversions in this context.

End of Paper